

You should work individually *at first*. After you have given the problems a good amount of independent thought, then you are encouraged to collaborate and work together with your Math 102 peers. **You are not allowed to look up solutions in any written form.** Write up your own solutions, and indicate any collaborators you worked with.

1. Consider the series  $\sum_{n=1}^{\infty} \frac{1}{7^n}$ .

(a) Use a calculator to calculate  $\sum_{n=1}^5 \frac{1}{7^n}$ .

(b) Use the integral test to estimate the error in using the partial sum  $S_5$  to approximate  $\sum_{n=1}^{\infty} \frac{1}{7^n}$ .

(c) Use the fact that  $\sum_{n=1}^{\infty} \frac{1}{7^n}$  is a geometric series to calculate the exact remainder term  $R_5$  in estimating with  $S_5$ . How does it compare to your answer in part (b)?

2. Does the series  $\sum_{n=1}^{\infty} \frac{\sin(n\pi/6)}{1 + n\sqrt{n}}$  converge conditionally, absolutely, or not at all?

3. Does the series  $\sum_{n=1}^{\infty} \frac{(-1)^n n^2 + e^{-n}}{2n^2 - n + 1}$  converge conditionally, absolutely, or not at all?

4. Does the alternating series  $\frac{2}{1} - \frac{1}{1} + \frac{2}{2} - \frac{1}{2} + \frac{2}{3} - \frac{1}{3} + \frac{2}{4} - \frac{1}{4} + \cdots$  converge or diverge? Be careful!

5. Recall that the alternating harmonic series  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$  converges conditionally.

(a) Let  $s_n$  denote the  $n^{\text{th}}$  partial sum of the series. Find an  $n$  such that the  $n^{\text{th}}$  partial sum will be within 0.05 of the infinite sum. Justify your answer. For your  $n$ , will  $s_n$  be an underestimate or an overestimate of the value of the infinite sum? Calculate your  $s_n$  as a decimal using a calculator or computer.

(b) Now consider the series  $1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$ . This series has the same terms as the alternating harmonic series, but in a different order. Why does the Alternating Series Test not apply to this series?

(c) It is a fact that this new series converges. Rewrite the series by simplifying the terms inside the parentheses:

$$\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \cdots$$

$$\underline{\hspace{1cm}} - \underline{\hspace{1cm}} + \underline{\hspace{1cm}} - \underline{\hspace{1cm}} + \underline{\hspace{1cm}} - \underline{\hspace{1cm}} + \cdots$$

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(d) Use your answer to find a constant  $M$  such that

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots = M \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

(**Remark:**  $M \neq 1$ , so rearranging the terms in an alternating series can change the value of the series!)